

Functional Features Overview

Date	5 July 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview

This document summarizes the functional features implemented in the **Smart Agricultural Production** project. The application is designed to assist farmers by recommending suitable crops based on soil and environmental parameters using a machine learning model. Each feature contributes to improving the usability, accuracy, and effectiveness of the system.

Functional Features

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Home Page	Provides the main interface for users to access the application.	User Interface	Done	High
2	Crop Recommendation	Predicts the most suitable crop based on user inputs.	Machine Learning Model	Done	High
3	Soil Parameter Input	Accepts soil nutrient values and environmental data from users.	Input Module	Done	High
4	Prediction Result Display	Displays the recommended crop with prediction results.	Output Module	Done	Medium
5	Dataset Integration	Uses agricultural datasets to train and support predictions.	Data Module	Done	Medium
6	Responsive User Interface	Ensures the application is usable on different screen sizes.	Frontend	Done	Medium

7	Input Validation	Validates user input before processing predictions.	Backend	Done	Medium
8	Flask Backend Processing	Handles requests, model execution, and response generation.	Backend	Done	High

Feature Summary

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Crop Recommendation, Soil Parameter Input, Prediction System, Flask Backend
Additional / Nice-to-Have Features	Responsive UI, Input Validation, Dataset Integration
Features Tested & Verified	8 Features Successfully Tested

Feature Category Breakdown

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Home Page, Prediction Result Page
2	Backend / Logic	3	Flask Routing, Input Validation, Prediction Processing
3	Database / Storage	1	Agricultural Dataset Storage
4	API / Integration	1	Machine Learning Model Integration
5	Security / Authentication	1	Input Validation and Error Handling